Band Theory

沈威宇

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1 Band Theory (能帶理論) and Semiconductor

I Basics

i Electronic band structure or band structure

The electronic band structure (or simply band structure) of a solid describes the range of energy levels that electrons may or may not have within it.

ii (Energy) band (能帶)

Atomic orbitals from multiple atoms combine to form molecular orbitals, such as bonding and anti-bonding orbitals. When a vast number of atoms—typically on the order of 10^{20} or more—come together to form a solid, the energy differences between these molecular orbitals become extremely small. Consequently, a cluster of closely spaced energy levels can be treated as a continuous energy band.

iii Fermi level (費米能階)

The Fermi level of a solid-state body is the thermodynamic work required to add one electron to the body. For Cu, $E_F = 7.0$ eV.

iv Work function (功函數)

For a material with Fermi level E_F and energy of an electron infinitely far in vacuum E_{vac}, the work function is

$$W = E_{\text{vac}} - E_F.$$

Work function is related with the threshold frequency f_0 of photoelectric effect by

$$W = hf_0.$$

v Density or concentration of carriers

For a type of charge carrier, the (number) density or concentration of it is defined as the number of it per unit volume in a material.

II Metals

i Bands

A partially filled band always exist, in which electrons can wander around and conduct current, called free electrons (or simply electrons).

ii Density of free electrons in partially filled band

Density of electrons in partially filled band n is the material's mass density multiplied by the material's number of valence electrons per atom multiplied by Avogadro's number divided by the material's molar mass.

iii Number density of states of the allowed energy level per unit volume per unit energy interval

The number density of states of the allowed energy level per unit volume per unit energy interval $N(E)$ is

$$N(E) = \frac{8\sqrt{2}\pi m_e^{3/2}}{h^3}\sqrt{E}.$$

iv Occupancy probability

The occupancy probability $P(E)$ is the probability that a given available state will be occupied by an electron

$$P(E) = \frac{1}{e^{\frac{E-E_F}{kT}} + 1}$$

where T is temperature in K and k is Boltzmann's constant 8.62×10^{-5} eV/K.

v Number density of occupied states per unit volume per unit energy interval

The number density of occupied states per unit volume per unit energy interval $N_0(E)$ is

$$N_0(E) = N(E)P(E).$$

vi Finding Fermi energy

E_F can be found by solving

$$n = \int_0^{E_F} N_0(E)\,\mathrm{d}E\,.$$

The result is

$$E_F = \left(\frac{3}{16\sqrt{2}\pi}\right)^{2/3} \frac{h^2}{m} n^{2/3} \approx \frac{0.121h^2}{m} n^{2/3}.$$

vii Thermal effect

The change of Fermi level on temperature change is small and often negligible for metals.

III Semiconductors

i Bands

For semiconductors and insulators,

- **Conduction band (導帶)**: The higher-energy band that is empty or slightly filled, in which electrons, if exist, can wander around and conduct current, called free electrons (or simply electrons). The energy of conduction band minimum is denoted E_C.
- **Valence band (價帶)**: The lower-energy, filled band, in which electrons are constrained in covalent bonds and can't wander away freely or conduct current. The energy of valence band maximum is denoted E_V.

- **Band gap (能隙)**: The gap between conduction band minimum and valence band maximum. The energy difference of the gap is called band-gap energy $E_g = E_C - E_V$. For silicon, $E_g = 1.12$ eV.

ii Insulator

A material with empty conduction band and high energy band such that virtually no transition of electron from valence band to conduction band occurs.

iii Semiconductor

A material with lower energy band such that transition of electron from valence band to conduction band can occur under room temperature and conductivity lies between that of conductors and insulators.

There are two kinds of semiconductors: single-element semiconductors, such as silicon (IV) and germanium (IV), and compound semiconductors, such as gallium arsenide GaAs (III-V), aluminium gallium arsenide AlGaAs (III-V), indium phosphide InP (III-V), Aluminium gallium indium phosphide AlInGaP (III-V), zinc oxide ZnO_2 (II-VI), and titanium doxide TiO_2 (II-VI). Compound semiconductors are useful in special electronic circuit applications as well as in applications that involve light, such as light-emitting diodes (LEDs). Germanium was used in the fabrication of very early transistors (late 1940s, early 1950s). It was quickly supplanted by silicon, on which today's integrated-circuit technology is almost entirely based. For this reason, we will deal mostly with silicon devices.

iv Energy and effective mass (有效質量) of electrons

The energy of an electron in a semiconductor is not the simple square relation of that of a free electron and wave number but a complicated periodic function. However, near an edge of a band, it can be approximated with:

$$E(k) \approx E_0 + \frac{\hbar^2 k^2}{2m^*}$$

where E_0 is the energy of the edge, k is wave number, and m[8] is a constant called effective mass given by

$$m^* = \hbar^2 \left(\frac{\mathrm{d}^2 E(k))}{\mathrm{d}k^2} \right)^{-1}$$

where $\frac{\mathrm{d}^2 E(k)}{\mathrm{d}k^2}$ is taken at the edge. Effective mass is usually written as multiple of mass of electron.

v Effective mass of electrons at difference levels

- At conduction band minimum, $\frac{\mathrm{d}^2 E(k)}{\mathrm{d}k^2} > 0$ and $m^* > 0$. These electrons behave like free electrons but with mass m^*.
- At valence band maximum, $\frac{\mathrm{d}^2 E(k)}{\mathrm{d}k^2} < 0$ and $m^* < 0$. These electrons move to the opposite direction of a force acting on it.

vi (Electron) hole (（電子）電洞/空穴)

A hole h^+ is an absence of an electron in valence band.

Let the effective mass of electrons at valence band maximum be $-m^*$ and the concentration of holes be p. Consider a nearly-filled valence band, the current of it equals to the current of a filled valence band, which is always zero, subtracting the current that would exist due to the electrons that would fill the holes if there are not the holes, which is in the opposite direction of the force. Thus, the current is in the direction of the force and equals to that due to particles with charge e and mass m^* at concentration p.

Thus, we define hole as a quasiparticle with charge e and effective mass m^*, which is positive.

vii Carriers (載子) of semiconductors

Thus, there are two types of carriers in semiconductors mathematically:

- free electron at conduction band minimum with charge $-e$ and a positive effective mass m_n^*, and whose properties are denoted with subscript n, and
- holes at valence band maximum with charge e and a positive effective mass m_p^*, and whose properties are denoted with subscript p.

The concentration of free electrons is denoted n; the concentration of holes is denoted p.

viii Effective densities of states

Available electronic states effectively participate in carrier occupation near a band edge at a given temperature, denoted as N_C for conduction band minimum and N_V for valence band maximum. They are governed by

$$N_C = 2\left(\frac{2\pi m_n^* kT}{h^2}\right)^{3/2}$$

$$N_V = 2\left(\frac{2\pi m_p^* kT}{h^2}\right)^{3/2}$$

ix Carrier generation

When an electron transits from valence band to conduction band, a free electron in conduction band and a hole in valence band are generated, and an energy of band-gap energy E_g is consumed.

Since they are generated in pairs, an intrinsic (undoped) semiconductor has equal concentration of free electrons and holes.

x Carrier recombination (載子複合) and electroluminescence (電致發光/電激發光)

When an electron transits from conduction band to valence band, a free electron in conduction band and a hole in valence band are cancelled, and an energy of band-gap energy E_g is released

in the form of a photon with frequency $f_h = \frac{E_g}{h}$, called electroluminescence.

Carrier recombination occurs naturally and at a rate proportional to the number of free electrons and holes.

Semiconductors with different band-gap energy can release light of different frequency. This effect is used in light emitting diode (LED) (發光二極體).

xi Exciton (激子)

A quasiparticle with no charge of a pair of free electron and hole in bounded state due to electrostatic force.

xii Photovoltaic effect (光生伏打效應/光伏效應)/Internal photoelectric effect (內光電效應)

Carrier generation caused by light.

Similar to the photoelectric effect, if the frequency of the incident light is greater than or equal to the material's threshold frequency $f_h = \frac{E_g}{h}$, electron-hole pairs are generated immediately; if the frequency of the incident light is less than this threshold, such pairs are never produced.

This effect is used in solar cell, photoresistor, and light-controlled relay.

xiii Thermal generation

Carrier generation due to thermal energy. The higher the temperature, the more thermal generations occur.

xiv Thermal equilibrium

Thermal equilibrium is the state where thermal generation rate equals to carrier recombination rate, at which

$$np = n_i^2$$

where n_i denotes the number of free electrons and holes in a unit volume at a given temperature given by

$$n_i = BT^{3/2}e^{-\frac{E_g}{2kT}}$$

where B is a material-dependent parameter that is $7.3 \times 10^{15}\ \text{cm}^{-3}\text{K}^{-\frac{3}{2}}$ for silicon, and

$$n = N_C \exp\left(-\frac{E_C - E_F}{kT}\right)$$

$$p = N_V \exp\left(-\frac{E_F - E_V}{kT}\right)$$

For intrinsic semiconductor at thermal equilibrium,

$$n = p = n_i$$

$$E_F = \frac{E_C + E_V}{2} + \frac{kT}{2}\ln\left(\frac{N_V}{N_C}\right) = \frac{E_C + E_V}{2} + \frac{3kT}{4}\ln\left(\frac{m_p^*}{m_n^*}\right)$$

At 0 K, no thermal generation occurs, so an intrinsic semiconductor has no carrier and behaves as an insulator at thermal equilibrium. At room temperature, the carrier concentration at thermal equilibrium is still far too small for intrinsic silicon to conduct appreciable current.

xv Doping (摻雜)

Doping is the intentional introduction of impurities into an intrinsic/undoped/pure semiconductor in order to change its properties. The doped material is referred to as an extrinsic/doped semiconductor.

xvi n-typed (n 型) semiconductors

Semiconductor with doping of elements that readily lose electrons, called donors, typically elements with valence of 5, such as phosphorus P and arsenic As, which introduces a substantial increase of the concentration of free electrons, Fermi level, and conductivity.

If the concentration of donor atoms is N_D, which is usually far greater than n_i, the concentration of free electrons in the n-type semiconductor will be approximately

$$n_n = N_D$$

and, at thermal equilibrium,

$$p_n = \frac{n_i^2}{N_D}$$

Hence, free electrons are said to be the majority (charge) carriers, while holes are said to be the minority (charge) carriers.

xvii p-typed (p 型) semiconductors

Semiconductor with doping of elements that readily get electrons, called acceptors, typically elements with valence of 3, such as boron B and aluminium Al, which introduces a substantial increase of the concentration of holes and conductivity and decrease of Fermi level.

If the concentration of acceptor atoms is N_A, which is usually far greater than n_i, the concentration of holes in the p-type semiconductor will be approximately

$$p_p = N_A$$

and, at thermal equilibrium,

$$n_p = \frac{n_i^2}{N_A}$$

Hence, holes are said to be the majority (charge) carriers, while free electrons are said to be the minority (charge) carriers.

IV Current flow in semiconductors

i Drude model of drift velocity and drift current

In Drude model, after a typical collision, each free electron or hole completely lose its memory of its previous velocity and moving off in a random direction. In the average time τ_n, τ_p between collisions, the average free electron and hole will acquire a drift speed of

$$\mathbf{v}_n = -\frac{e\tau_n}{m_n^*}\mathbf{E}$$

$$\mathbf{v}_p = \frac{e\tau_p}{m_p^*}\mathbf{E}$$

We define the free electron mobility μ_n and hole mobility μ_p as

$$\mu_n = \frac{e\tau_n}{m_n^*}$$

$$\mu_p = \frac{e\tau_p}{m_p^*}$$

For silicon, μ_n is 1350 $\text{cm}^2\text{V}^{-1}\text{s}^{-1}$ and μ_p is 480 $\text{cm}^2\text{V}^{-1}\text{s}^{-1}$.

The charge densities are

$$d_n = -ne$$

$$d_p = pe$$

The drift current densities are

$$\mathbf{J}_{d,n} = d_n\mathbf{v}_n = -ne\mathbf{v}_n = ne\mu_n\mathbf{E} = \frac{ne^2\tau_n}{m_n^*}\mathbf{E}$$

$$\mathbf{J}_{d,p} = d_p\mathbf{v}_p = pe\mathbf{v}_n = pe\mu_p\mathbf{E} = \frac{pe^2\tau_p}{m_p^*}\mathbf{E}$$

The total drift current density is

$$\mathbf{J}_d = \mathbf{J}_{d,n} + \mathbf{J}_{p,n} = e\left(n\mu_n + p\mu_p\right)\mathbf{E} = e^2\left(\frac{n\tau_n}{m_n^*} + \frac{p\tau_p}{m_p^*}\right)\mathbf{E}$$

Thus the resistivity ρ is

$$\rho = \frac{|\mathbf{E}|}{|\mathbf{J}|} = \frac{1}{e\left(n\mu_n + p\mu_p\right)} = \frac{1}{e^2\left(\frac{n\tau_n}{m_n^*} + \frac{p\tau_p}{m_p^*}\right)}$$

which is a constant, i.e., obeying Ohm's law.

ii Diffusion current

When the concentration of a type of charge carriers differ across a piece of semiconductor, the charge carriers will diffuse from he region of high concentration to the region of low concentration, which gives rise to a net flow of charge, called diffusion current.

The concentration of charge carriers of a piece of semiconductor under a state is called concentration profile.

The diffusion current densities of free electrons and holes are

$$\mathbf{J}_{D,n} = -eD_n \nabla n$$

$$\mathbf{J}_{D,p} = eD_p \nabla p$$

where D_n, D_p are constants called diffusion constants or diffusivities of free electrons and holes respectively given by:

$$D_n = \mu_n V_T$$

$$D_p = \mu_p V_T$$

called Einstein relationship, where V_T is thermal voltage given by

$$V_T = \frac{kT}{e}$$

For $T \approx 300$ K, V_T is 25.9 mV. For intrinsic silicon, D_n is 35 cm^2/s and D_p is 12 cm^2/s.

V p–n junction (p–n 接面)

i Definition

A p-n junction is a single semiconducting crystal with one end doped to form p-type material and the other end doped to form n-type material. The two types meet at a junction plane. At thermal equilibrium, the following occur at the junction plane:

- Majority carriers diffuse across the plane, producing a diffusion current.
- Minority carriers are swept across the plane, forming a drift current.
- A depletion zone forms at the plane. It is normally a region of high resistivity, but can be decreased by decreasing its width.
- A contact potential develops across the depletion zone. The larger it is, the less the diffusion current is. The drift current is not affected by it.

-

ii Junction rectifier

A p-n junction conducts electricity better for one direction of an applied potential difference (forward biased) than for the opposite direction (back biased), and thus the device can serve as a junction rectifier.

In forward-bias connection, the p side becomes more positive and the n side becomes more negative, thus decreasing the height of the potential barrier, increasing the diffusion current, narrowing the depletion zone, and decreasing its resistivity.

In back-bias connection, the p side becomes less positive and the n side becomes less negative, thus increasing the height of the potential barrier, decreasing the diffusion current, widening the depletion zone, and increasing its resistivity.

iii 發光二極體（Light emitting diode, LED）

將順向偏壓通入半導體二極體，使 depletion zone 極窄，使其中大量發生載子複合而電致發光。The frequency f of the emitted light is

$$E_g = hf.$$

早期研發出的紅色和綠色單色發光二極體用途較受限，1992 年赤崎勇、天野浩與中村修二以氮化鎵材料發明藍光 LED，始可用光的三原色調配任意所需的顏色，尤其用作照明的白光，較傳統白熾燈泡與日光燈更省電、體積更小、壽命更長，共同獲 2014 年諾貝爾物理獎。

iv 有機發光二極體（Organic light-emitting diode, OLED）

有機半導體製成的二極體，原料如「三 (8-羥基喹啉) 鋁/三 [8-羥基 (苯 [b] 氮雜環己-1,3,5-三烯)] 鋁（tris(8-hydroxyquinolinato) aluminium(III)）$Al(C_9H_6NO)_3$」或「聚對苯乙炔（Poly(p-phenylene vinylene), PPV）[-c1ccc(cc1)C=C-]$_n$」。

v Photodiode

Shining light on a suitably arranged p-n junction can that includes the junction. This is the basis for the photodiode.

vi Junction laser

In a forward-bias connection connected to a source, there are many electrons in the conduction band of the n-type material and many holes in the valence band of the p-type material. Thus, there is a population inversion for the electrons; that is, there are more electrons in higher energy levels than in lower energy levels. When a single electron moves from the conduction band to the valence band, it can release its energy as a photon. This photon can stimulate a second electron to fall into the valence band, producing a second photon by stimulated emission. In this way, if the current through the junction is great enough, a chain reaction of stimulated emission events can occur and laser light can be generated. To bring this about, opposite faces of the p-n junction crystal must be flat and parallel, so that light can be reflected back and forth within the crystal. Junction lasers are built into compact disc (CD) players, where, by detecting reflections from the rotating disc, they are used to translate microscopic pits in the disc into sound. They are also much used in optical communication systems based on optical fibers.

vii 光伏電池（Photovoltaic cell）/太陽能電池（Solar cell）

通常由一個或多個 p–n 接面組成，吸收光子並輸出直流電。種類如：

- 單結晶矽太陽能電池：高效率但成本較高。
- 多晶矽太陽能電池：效率較低但成本較低。
- 薄膜太陽能電池：輕薄柔軟，適用於特定場景。

效率提升方法：

- 減少表面反射：透明導電層和抗反射層減少光子的反射。
- 改進材料：新材料如有機太陽能電池和混合鈣鈦礦太陽能電池提高效率。
- 多接面結構：提高光吸收和轉換效率。
- 雷射細工：提高電池表面光陽極的粗糙度，增加光吸收。

viii 場效電晶體（Field-effect transistor, FET）

FET 具有四個端子：源極（source）、閘極（gate）和汲極（drain）、體（body）/基（base）/塊體（bulk）/基板（substrate）。通過在閘極施加電壓，可以改變源極與汲極之間的電導率，從而控制電流，分為需加閘極電壓方能導通源極與汲極的增強模式（enhancement-mode），較常見；與預設導通源極與汲極，加閘極電壓可減弱或關閉它的空乏模式/減少模式（depletion-mode）。

下討論 metal-oxide-semiconductor-field-effect transistor (MOSFET)。其僅使用電子與電洞的其中一種作為電荷載子，前者稱 n-通道（n-channel），後者稱 p-通道（p-channel）。源極與汲極下方的摻雜半導體具有該種載子，在增強模式中預設不相連而斷路，但加閘極電壓可以引入載子使相連而導通；在空乏模式中預設相連而導通，但加閘極電壓可以引入相反的載子使變窄增加電阻，繼而不相連而斷路。